SATURN: Splitting Approach on Tensorized Ubiquitous Regression Networks

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Abstract

Convolutional Neural Networks often contain fully connected layers after their convolution phase, which requires the data to be flattened. Despite empirical success, this flattening ruins the multidimensional structure of the input data and increases the number of parameters, which may cause overfitting. Given the tensor nature of the input data, fully connected layers can be replaced with tensorized low-rank layers, which, besides considering the nature of the data, reduce the number of parameters. However, these layers perform poorly when the approximated rank is too low. To address this issue and further reduce the number of parameters while preserving the quality of the network, we propose a framework that, by using two splitting methods, transforms the multidimensional tensor into a thinner and higher-order tensor and applies tensorized layers on this high-order tensor. This leads to fewer parameters and less computational cost while preserving the quality of the network. In our experiments, evaluated on CIFAR10, MNIST, and Tiny ImageNet-200, we have successfully reduced the classifier parameters of well-known models such as VGG19, ResNet50, and ResNet101 by more than 95%, while also achieving better accuracy and overcoming overfitting. Compared to traditional state-of-the-art models, our approach achieves significant parameter reduction and compression while preserving accuracy.

1 Introduction

Neural Networks, especially those that are applied to images such as *Convolutional Neural Network* (CNNs), often consist of two main parts: One that is responsible to extract features from the input data, and the other part is a *MultiLayer Perceptron* (MLP) which Maps the extracted features to the desired space. Despite the importance of data and their spatial relationship, maintaining their topological structure is often discarded since many known CNNs like ResNet or VGG, seem to work fine; However in cases where these structures are more important than the others such as MRI, It is crucial to adapt the known networks to work without destroying the multi-linear structure of the data. While preserving the natural structure of the data throughout the network is important, a greater drawback from using MLPs is their excessive required number of parameters which not only limit their capabilities in real time and practical problems but also prone them to overfitting.

Most natural datasets can be represented as a Tensor due their multi-linear or multi-modal structure; Images can be considered as a 3-order tensors with width, height and the color channels as their corresponding 3 modes, the same could be considered for videos as 4-order tensor with an extra time mode. In practice, most of the used data in the current machine learning and artificial intelligence tasks can be encoded as tensors, requiring their expansion to the linear algebra.

In machine learning, we usually consider data as vectors and datasets will form a set of vectors (ideally called a matrix). By that definition, many matrix decomposition methods have been invented

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to work on such matrices. Tensor methods are to generalize these methods to a higher dimensional space. Mathematically, tensors, have been widely studied in the past. One excellent example within tensor methods is Tensor Decomposition which can be considered as a prime backbone for most tensorized layers and approximations in deep learning [1] [2], Signal Processing [3] [4], fusion [5], blind source seperation [4] [6], and computer vision [7] .

Considering the new advances in *Deep Neural Networks* (DNNs) and their applications to computer vision such as image recognition, image segmentation, image generation, video generation, vision transformers, etc ...; most architectures still flatten the data before feeding them to their MLPs which not only destroys the natural structure of the data but also increase the number of parameters needed for the model.

Images as the frequent example of data used in machine learning tasks are considered 3-order tensors which require their own algebra to be applied when fed to the MLP. *Fully Connected* (FC) layers used in many architectures are the main culprit behind vectorizing the data and the number of parameters. To alleviate this situation, a better replacement for FC layers is needed which not only preserves the current model's accuracy but also reduces the number of required parameters and does not destroy the structure of data.

One possible working solution to tensorizing FC layers is by employing the low rank approximation of the weight tensor. Some works have been done to address these issues which will be discussed in Section 2 . However, those methods usually perform poorly when the approximated rank is too low which is not ideal for larger models. The cost of these models usually depend on the size of the tensor, our experiment shows that a thinner while higher dimensional reshaped tensor of the original input tensor, can perform much better in lower ranks and by a unique splitting and stacking technique, we preserved the models' accuracy. In this paper, we have provided two splitting approaches which have their own use dependent on the model's task; Using these methods, we have reshaped the tensor to a higher dimensional space where the cost of computation is less and has a better capability to handle lower approximated ranks. Our methods evaluated on CIFAR10, MNIST, and Tiny-ImageNet-200 applied to known CNN models such as ResNet50, ResNet101, and VGG19, as well as a typical CNN achieved similar accuracy in most methods with more than 95 percent less parameters compared to the FC layers and even overcame overfitting in some cases.

2 Related works

Several prior papers address the importance of using tensors and preserving the natural structure, One uses CP decomposition to enhance convolution layers [8]. In [9], tucker decomposition is used for convolution enhancement with fewer parameters. A weight sharing scheme in multi tasking learning is proposed by [10]. In [11], a Tensor-Train decomposition is used as a low rank approximation of the weight tensor. Some uses tensor methods for model compression and acceleration [12].

In general, tensor methods can be widely used in many machine learning and deep learning tasks, especially computer vision [13]. Their use, can be categorized into three main approaches: The first approach compresses the tuned parameters of the pre-trained network while preserving the quality of network with these approximated parameters. So here the new network is not trained and the approximated network could only be used in the test step. Two methods for approximation of the tuned parameters of pertained CNN in each layer based on matrix and tensor decompositions adjusted with clustering of inlet and outlet layers have been proposed [14]. An improvement is reported in [15]. Tensor CP and Tucker decompositions are used to approximate the kernels of pre-trained CNN networks [16] [17] [18]. In [18] a tensor based method to find an approximation of fully connected layer parameters in CNN networks is introduced. Despite the proper implementation results, these methods have the following drawbacks: increase cost of pre-trained networks and finding these compressions is also costly. It also cannot be used in applications such as deep reinforcement learning where the network constantly should be learned. On the other hand, this approach has confined itself to compressing convolutional methods for the test phase, although a deeper question can be asked. Why should we compress an already trained network and not seek for a compact network representation that can be trained from scratch?

The second approach consider the convolutional layers as a base, and approximate or represent them with smaller parameters, to design end-to-end trainable neural network. Different kinds of methods

have been proposed to design end-end learnable networks based on approximation of convolution process with different views: For example, in [19] and [20] , the rank-1 approximation of 3-order tensor filters corresponding to each outlet channel is used. In [21] ,a share factors corresponding to inlet and outlet are considered to reduce the redundancy of parameters. Recently, in [22] by considering the 3-order filters of each outlet channel as rank-k tensor it has been shown that the pixels of each outlet channel is the contraction of this tensor with the corresponding patch of all input channels. Unlike standard convolution, Mobile Nets proposed in [23] separate the spatial filtering and integration steps. For each input channel, only a single spatial filtering is performed, and by different combination of these filtered images, different output channels are obtained.

The third approach do not consider Convolutional structure as a base and try to find other type of layers to work on multidimensional data. One notable recent work is Tensor Regression Networks [1]. They used a *Tensor Regression Layer* (TRL) and *Tensor Contraction Layer* (TCL) to preserve the structure of the data as they are moving across the network while reducing the number of parameters. Their method surprisingly achieved the near similar accuracy as its FC network with more than 65 percent space saving. TRL and TCL will be discussed in section 3, and they are used as the base of our method. Unfortunately, one of the biggest issues with TCL and TRL is their inadequate implementations. They are theoretically required less parameters than their FC versions; However they often perform poorly in terms of the actual execution time. This is one of the biggest limitation on working with tensors since a practical implementation of tensor methods are yet to be made.

The power of tensor regression has been addressed in many prior papers [24][25] [26] [27]. However, most of these works, usually rely on analytical solution and large tensors. In [28], a non-linear Tensor-Train decomposition is used instead of tucker decomposition. While many works focuses on individual layers, in [29], T-Nets attempts to provide a single high-order, low-rank tensor that jointly capture the full structure of a neural network.

DNNS have been used in many applications and problems, however, many questions still remains in terms of their credibility and why they work so well. DNNs usually contain too many parameters and an important question is whether they require that many parameters. By tensorizing the DNNs, we can address these issues and attempt to better understand the concepts of deep learning. For example, one uses tensor methods as a tool to study CNNs [30], they follow up by studying the power of overlapping architectures in deep learning [31]. Some provide global optimal and optimization for non-convex factorization problems [32]. In [33], they used contraction as a composition operator in *Natural Language Processing* (NLP). Some used tensor methods as a tool to guarantee convergence and consistent learning[34].

Most of the recent works, rely on low rank approximation of the tensor or some tensor decomposition methods as a dimension reduction layer. Despite their success, a fully tensorized end-to-end trainable model requires almost near similar ranks to perform well when compared to their FC competitives. The draw back of using same ranks as the input tensor is the cost. Our method provide a reshaping approach that maintains the accuracy by converting the input tensor into a thinner and higher order tensor that requires fewer parameters and can perform better in lower ranks. To our knowledge, no prior work increases the dimensions of the tensor in order to achieve near similar accuracy while using tensorized layers with lower ranks and fewer parameters.

3 Mathematical background

This paper, requires basic understanding of tensor methods, If readers are already knowledgeable in the matter, they can skip to section 4.

3.1 Notations

For a 3-order tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, there are 3 modes with respected sizes I_1, I_2, and I_3. Its $\mathcal{T}_{i_1,i_2,i_3}$ **element** is a 0-order tensor (scalar), (i_1, i_2, i_3). **Slices** of its first and second modes are 2-order tensors (matrices) $\mathbf{M}$ and are denoted by two colons, $\mathcal{T}_{:,:,i_3}$. **Fibers** of its first mode are 1-order tensors (vectors) $\mathbf{V}$ and are denoted by a colon, $\mathcal{T}_{:,i_2,i_3}$.

3.2 Tensor folding and unfolding

Tensor folding and unfolding are fundamental operations in tensor analysis, allowing for the manipulation and transformation of tensors into different dimensional structures. Tensor unfolding, also known as matricization, converts a higher-order tensor into a matrix or a vector. Let $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \ldots \times I_N}$ be an N-order tensor. In our case, tensor folding is the inverse operation of tensor unfolding, which reconstructs a higher-order tensor from a matrix or vector.

Mode-n unfolding : Unfolding $\mathcal{T}$ along mode-n [2] results in a matrix representation of mode-n. This operation is denoted as $T_{[n]}$. The unfolding is achieved by rearranging the entries of $\mathcal{T}$ into a matrix such that the mode-n indices become the rows of the matrix. The mapping from tensor indices to matrix indices is given by:

$$(i_1, \ldots, i_n, \ldots, i_N) \longrightarrow (i_n, r) \qquad r = \sum_{\substack{p=1 \\ p \neq n}}^{N} i_p \times \prod_{\substack{l=p+1 \\ l \neq n}}^{N} I_l \tag{1}$$

Each row of the resulting matrix corresponds to a different combination of indices for the remaining modes.

Vectorization : Similar to mode-n unfolding, in tensor vectorization [2] we map each element within the tensor to a element j of $Vec(\mathcal{T})$:

$$(i_1, \ldots, i_n, \ldots, i_N) \longrightarrow (r) \qquad r = \sum_{p=1}^{N} i_p \times \prod_{l=p+1}^{N} I_l \tag{2}$$

3.3 Tensor contraction

Tensor contraction allows for the multiplication and simplification of tensors and provides a way to combine tensors by summing over shared indices.

Tensor n-mode product : Tensor n-mode product [2] , is a tensor operation that allows for the multiplication of a tensor with a matrix along a specific mode. Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and a matrix $\mathbf{M} \in \mathbb{R}^{J \times I_n}$, the n-mode product of $\mathcal{T}$ and $\mathbf{M}$, denoted as $\mathcal{T} \times_n \mathbf{M}$, results in a tensor of size $\mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$. The n-mode product can be expressed mathematically as:

$$(\mathcal{T} \times_n \mathbf{M})_{i_1, \ldots, i_{n-1}, j, i_{n+1}, \ldots, i_N} = \sum_{i_n=0}^{I_n} \mathcal{T}_{i_1, \ldots, i_n, \ldots, i_N} \mathbf{M}_{j, i_n} \tag{3}$$

where $i_1, \ldots, i_{n-1}, j, i_{n+1}, \ldots, i_N$ are the indices of the resulting tensor, and i_n is the summation index. Alternatively, the n-mode product can be expressed using the tensor unfolding operation along the n-th mode, denoted as $\mathbf{T}_{(n)}$. In this case, the n-mode product can be written as:

$$\mathcal{T} \times_n \mathbf{M} = \mathbf{M}\mathbf{T}_{[n]} \tag{4}$$

Generalized inner product : For two tensors $\mathcal{X} \in \mathbb{R}^{I_x \times I_1 \times I_2 \times \cdots \times I_N}$ and $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N \times I_y}$ sharing N modes of the same size, a generalized inner product [2] along the N last modes of X and Y is defined as:

$$\langle \mathcal{X}, \mathcal{Y} \rangle_N = \sum_{i_1=0}^{I_1-1} \cdots \sum_{i_N=0}^{I_N-1} \mathcal{X}_{:, i_1, i_2, \ldots, i_N} \mathcal{Y}_{i_1, i_2, \ldots, i_N, :} \tag{5}$$

where $\langle \mathcal{X}, \mathcal{Y} \rangle_N \in \mathbb{R}^{I_x \times I_y}$.

3.4 Tucker decomposition

The Tucker decomposition [2] is a powerful method in tensor algebra that breaks down a tensor into a core tensor multiplied by matrices along each mode. In the Tucker decomposition, a tensor $\mathcal{T} \in$

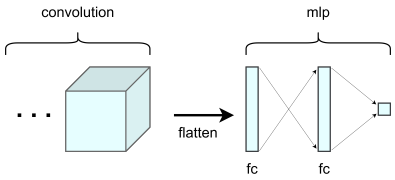


Figure 1: A simple MLP after some convolution layers.

$\mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is represented as the product of a core tensor $\mathcal{G} \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_N}$ and a set of factor matrices $\mathbf{A}^{(1)} \in \mathbb{R}^{J_1 \times I_1}, \mathbf{A}^{(2)} \in \mathbb{R}^{J_2 \times I_2}, \ldots, \mathbf{A}^{(N)} \in \mathbb{R}^{J_N \times I_N}$ along each mode. Mathematically, the Tucker decomposition can be expressed as:

$$\mathcal{T} = \mathcal{G} \times_1 \mathbf{A}^{(1)} \times_2 \mathbf{A}^{(2)} \times \cdots \times_N \mathbf{A}^{(N)} \tag{6}$$

3.5 Tensor regression networks

The combination of tensor regression and tensor contractions has been used in an end-to-end trainable model by Kossaifi [1], as discussed in section 2 Given a MLP, there are two ways to interpret FC layers: One is to consider them as a feature extraction layer where they map a latent or feather space to another latent space. Another approach is to consider them as regression layers which are basically the classifier layers and given the purpose of the network, they can be used interchangeably. In figure 1 a very simple MLP that is being used after some convolution layers is provided. In this case, the first FC layer can be considered a dimension reduction layer while the second FC layer maps the feature space to the target space. In the following text, we will discuss two main approaches to replace such FC layers:

3.5.1 Tensor contraction layer

Tensor Contraction Layer (TCL) [1] is a form of feature extraction and could also be used for embedding. The prime idea behind TCL is Tucker decomposition; Via multiplying matrices to each mode of the input tensor, we can obtain a denser tensor . Figure 2 demonstrate the TCL process on a 3-order tensor.

Given an input tensor $\mathcal{X} \in \mathbb{R}^{S_0 \times I_1 \times \cdots \times I_N}$, a rank $(R_1, \ldots, R_N)$ TCL on $\mathcal{X}$ is defined as :

$$\begin{aligned} \hat{\mathcal{X}} &= \left(V^{(1)}, \ldots, V^{(N)}\right)_{1:N} \mathcal{X} \\ \hat{\mathcal{X}}_{[0]} &= \mathcal{X}_{[0]}(V^{(1)} \otimes \cdots \otimes V^{(N)})^T \end{aligned} \tag{7}$$

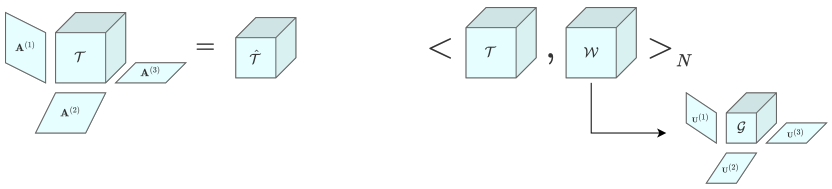
where S_0 is the batch index and $V^{(i)} \in \mathbb{R}^{R_i \times I_i}$ is the TCL factor; The number of required parameters is calculated as :

$$\sum_{i=1}^{N} R_i \times I_i \tag{8}$$

And the resulting tensor is of size $(S_0, R_1, \ldots, R_N)$. Compared to the FC version where the simple regression between these two tensors would require at least $\prod_{i=1}^{N} I_i \times R_i$ parameters.

3.5.2 Tensor regression layer

Tensor Regression Layers (TRL) [1] is a replacement for the FC regression layers. A simple regression that is the foundation of a MLP is a trivial generalized inner product of $\mathcal{W}$ (Weight Tensor) with the $\mathcal{X}$ (Input tensor). The idea behind TRL is to reconstruct W from a low rank approximation of itself via tucker decomposition. In figure 2, a TRL application in a MLP is provided. For the sake of simplicity, we considered a binary classification problem where the output of the inner product is a scaler. Similarly, if the desired output is a vector or even a tensor, $\mathcal{W}$ will be a higher dimensional tensor itself to generate each mode of the output.



(a) (b)

Figure 2: Tensor regression networks, replacement for FC layers. In (a), A simple TCL is applied; Each mode i of tensor $\mathcal{T}$ is multiplied by factor $\mathbf{A}^{(i)}$. In (b), A TRL is applied; Generalized inner product of input tensor $\mathcal{T}$ and weight tensor $\mathcal{W}$ is calculated as the output tensor while considering a low rank approximation of weight tensor $\mathcal{W}$ in a tucker decomposed form.

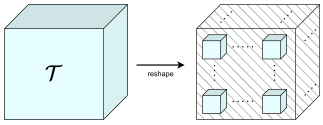


Figure 3: An outline of reshaping to higher orders tensor.

Given an input tensor $\mathcal{X} \in \mathbb{R}^{S_0 \times I_1 \times \cdots \times I_N}$, output tensor $\mathcal{Y} \in \mathbb{R}^{S_0 \times O}$, a rank $(R_1, \ldots, R_N, R_{N+1})$ TRL on $\mathcal{X}$ is defined as:

$$\begin{aligned} \mathcal{W} &= \left(U^{(1)}, \ldots, U^{(N)}, U^{(N+1)}\right)_{0:N+1} \mathcal{G} \\ \mathcal{Y} &= \langle \mathcal{X}, \mathcal{W} \rangle_N \end{aligned} \tag{9}$$

where S_0 is the batch index, $\forall i \; U^{(i)} \in \mathbb{R}^{R_i \times I_i}$, $U^{(N+1)} \in \mathbb{R}^{R_{N+1} \times O}$, and $\mathcal{G} \in \mathbb{R}^{R_1 \times \cdots \times R_N \times O}$; The number of required parameters can be calculated as :

$$\prod_{i=1}^{N+1} R_i + \left(\sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O \right) \tag{10}$$

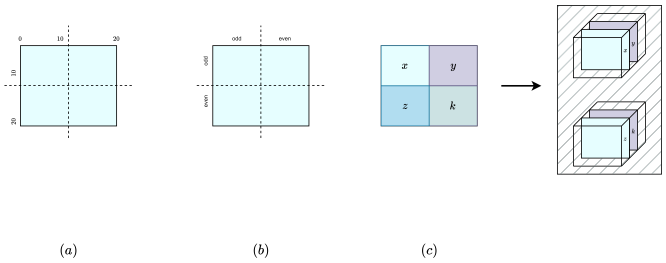
Compared to the FC version where the number of parameters is $O \times \prod_{i=1}^{N} I_i$.

4 Splitting approach

Impressed by the vision transformers patching and down sampling methods, we can transform our input tensor into a thinner and higher dimensional tensor that not only reduces the number of parameters for a TRL or TCL layer but also maintain the accuracy. However, not any reshaping scheme to arbitrary higher orders and dimensions guarantee that the multi-linear structure is preserved and the accuracy is maintained With a proper splitting scheme, the newly made tenser has better physical interpretation than the flattened data. We introduced two splitting schemes, both of which are useful and could be used differently according to the network architect. Figure 3 demonstrate the general splitting outline. Given a 3-order tensor, if we split across each mode, the output is a 6-order tensor where the first 3 modes correspond the split index, and the next 3 modes are the split data.

4.1 Attached splitting

In attached splitting, every split is continues, that is if mode I_i of the input tensor $\mathcal{X}$ with length n is to be split into 2 parts, then the split indices are $(0, \frac{n}{2})$ and $(\frac{n}{2}, n)$. Figure 4 demonstrate the splitting process on a matrix.



(a) (b) (c)

Figure 4: (a) attached splitting of a 2-order tensor. (b) compressed splitting of a 2-order tensor. (c) demonstration of stacking split chunks of a 2-order tensor into a 4-order tensor.

Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ and split sizes $(L_1, \ldots, L_N)$, attached mapping of elements is done via:

$$\mathcal{T}[i_1, \ldots, i_N] \longrightarrow \mathcal{T}_a \left[\lfloor \frac{L_1 \times i_1}{I_1} \rfloor, \ldots, \lfloor \frac{L_N \times i_N}{I_N} \rfloor, i_1 \bmod \frac{I_1}{L_1}, \ldots, i_N \bmod \frac{I_N}{L_N}, \right] \tag{11}$$

where $\mathcal{T}_a \in \mathbb{R}^{L_1 \times \cdots \times L_N \times \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}}$.

4.2 Compressed splitting

In compressed splitting, we incorporate the idea behind down sampling, that is if mode I_i of input tensor $\mathcal{X}$ with length n is to be split into 2 part , different to the attached splitting, the indices are even-odd combinations; That is $\{t | t \in (0, n) \wedge (t \bmod 2 = 0)\}$ and $\{t | t \in (0, n) \wedge (t \bmod 2 = 1)\}$.

Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ and split sizes $(L_1, \ldots, L_N)$, compressed mapping of elements is done via:

$$\mathcal{T}[i_1, \ldots, i_N] \longrightarrow \mathcal{T}_c \left[i_1 \bmod L_1, \ldots, i_N \bmod L_N, \lfloor \frac{i_1}{L_1} \rfloor, \ldots, \lfloor \frac{i_N}{L_N} \rfloor, \right] \tag{12}$$

where $\mathcal{T}_c \in \mathbb{R}^{L_1 \times \cdots \times L_N \times \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}}$.

4.3 Method analysis

Considering an input tensor $\mathcal{X}$ of size $(S_0, I_1, \ldots, I_N)$, either reshaping would result in the tensor $\hat{\mathcal{X}}$ of size $(S_0, L_1, \ldots, L_N, \frac{I_1}{L_1}, \ldots, \frac{I_N}{L_N})$. An example of each reshaping approach is done in figure 5 on a sample image.

In case of a $(r_1 \ldots, r_N, R_i, \ldots, R_N)$ TCL, Since L_is are usually small, the number of parameters required for them $\sum_{i=0}^{N} L_i \times r_i$ is also small; Similarly, $\frac{I_i}{L_i}$ is smaller than I_i so the new desired R_is is smaller than the previous R_is and the total number of parameters can be calculated as:

$$\sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} \tag{13}$$

which is ideally less than equation 8 (Appendix A and C).

In case of $(r_1, \ldots, r_N, R_1, \ldots, R_N, R_{N+1})$ TRL, the same logic as TCL could also be applied. The total number of parameters in a new TRL is calculated as:

$$\left(\prod_{i=1}^{N} r_i \prod_{i=1}^{N+1} R_i \right) + \left(\sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} + R_{N+1} \times O \right) \tag{14}$$

which is ideally less than equation 10 (Appendix B and D).

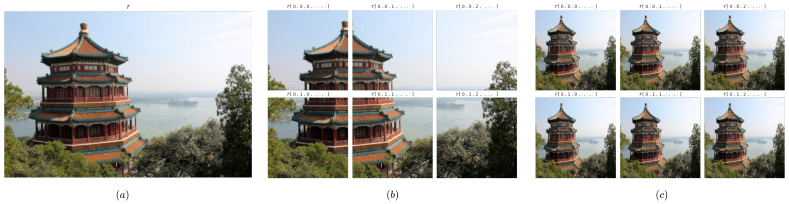


Figure 5: (a) original sample image. (b) reshaped sample image using attached approach. (c) reshaped sample image using compressed approach.

Table 1: **Performance evaluation of ResNet50 and ResNet101**, Our methods reduce the number of parameters by more than 90 % while achieving near similar accuracy. More detailed Table(s) at (Appendix E).

Dataset	Model	TRL	Split	Parameters	Top-1 (%)
MNIST	Resnet50	Baseline		737,290	97.25
		(50,3,3,3)		103,816	96.49
		(1,1,1,20,1,1,4)	$(2,2,2)^A$	20,612	98.56
		(4,1,1,20,1,1,10)	$(16,2,2)^C$	3,534	98.12
	Resnet101	Baseline		737,290	98.58
		(50,3,3,3)		103,816	97.42
		(1,1,1,20,1,1,4)	$(2,2,2)^C$	20,612	98.24
		(4,1,1,20,1,1,10)	$(16,2,2)^A$	3,534	98.40
CIFAR10	Resnet50	Baseline		737,290	90.71
		(50,3,3,3)		103,816	90.41
		(1,1,1,100,1,1,10)	$(2,2,2)^C$	103,512	92.58
		(4,1,1,10,1,1,4)	$(16,2,2)^A$	1,386	90.44
	Resnet101	Baseline		737,290	92.67
		(50,3,3,3)		103,816	90.30
		(1,1,1,100,1,1,10)	$(2,2,2)^C$	103,512	91.85
		(4,1,1,10,1,1,4)	$(16,2,2)^A$	1,386	92.43

5 Experiments

We conducted a series of experiments to assess the performance of our proposed framework.Our experiments were performed on GeForce RTX 4090; Datasetes MNIST, CIFAR10, Tiny ImageNet-200 applied to ResNet50, ResNet101, VGG-19, and classic CNN demonstrate that our methods can significantly reduce the number of parameters while maintaining or even improving classification accuracy. In Tables 1 , 2 , and 3 parameters include only the classifier parameters. For a better representation of our methods, we ignored adaptive average pooling layers in all models. ^{A} and ^{C} are splitting approaches. Our framework performs exceptionally well compared to traditional TCL TRL and baseline models across various datasets and architectures. Notably, our approach significantly

Table 2: **Tiny ImageNet-200 performance on VGG-19**, Given extreme number of classifier parameters for a VGG-19 model, we applied TCL layers before TRL layers. More detailed Table(s) at (Appendix E).

TCL	TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline			93,102,280	63.66	84.16
(512,6,6)	(512,6,6,200)		4,250,832	67.32	87.87
(2,2,2,256,1,1)	(1,1,1,256,1,1,200)	$(2,2,2)^A$	666,714	68.91	88.16
(6,2,2,64,3,3)	(4,2,2,64,1,1,200)	$(8,2,2)^C$	253,104	68.29	88.34

Table 3: **Classic CNN on MNIST**, Our methods are dependent on a good convolutional foundation. More detailed Table(s) at (Appendix E).

TCL	TRL	Split	Parameters	Top-1(%)
Baseline			4,729,866	98.14
(128,6,6)	(128,6,6,10)		79,092	95.62
(2,1,1,64,6,6)	(2,1,1,64,6,6,10)	$(2,1,1)^A$	54,528	97.11
(2,1,1,64,6,6)	(2,1,1,64,6,6,10)	$(2,1,1)^C$	54,528	97.12

reduces the number of parameters without acutely compromising accuracy. The **ablation study** is carried out in the Appendix F.

6 Conclusion

Deep neural networks, especially CNNs, often work with multi-dimensional tensors. Despite their empirical success, these architectures usually contain one or more fully connected layers which not only destroy the natural multi-linear structure of the data, but also over parameterize the model. By introducing tensorized layers such TRL and TCL we can achieve near similar accuracy while preserving the multi-modal structure of the data and reducing the number of classifier parameters; However, these methods perform poorly when the approximated rank is too low. We introduced SATURN, a framework which maintains the quality of the network and reduces the number of classifier parameter by more than 95 %. Our experiment, evaluated on CIFAR10, MNIST, and Tiny ImageNet-200, applied to RestNet50, RestNet101, VGG19 , and classic CNNs demonstrate that by splitting the input tensor via an approach that does not disturb the multi-linear structure of the data, a TCL or TRL with fewer parameters can be performed which also maintains the accuracy. Based on our observations, many popular CNNs learn their convolutional layers based on the flattening operation and their following FC layers. Therefore, for our framework to achieve best accuracy, all layers of the models need to be trained for at least a few iterations. While our framework and similar approaches offer theoretical advantages, they currently do not exhibit the expected speed. This limitation underscores the primary challenge in applying tensor methods to models, as the development of optimal implementations remains a pending task.

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A Method analysis on TCL

As stated in subsection 4.3, equation 13, the number of required parameters for a TCL on our framework is ideally less than equation 8. Given $\mathcal{T} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ and $\hat{\mathcal{T}} \in \mathbb{R}^{L_1 \times \cdots \times L_N \times \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}}$, ideally a rank $(r_1, \ldots, r_N, \hat{R}_1, \ldots, \hat{R}_N)$ TCL on $\hat{\mathcal{T}}$ has less parameters than a rank $(R_1, \ldots, R_N)$ TCL on $\mathcal{T}$. Since TCL ranks are arbitrary values, one can choose these ranks in a way that inequality 15 does not hold.

$$\sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} \leq \sum_{i=1}^{N} R_i \times I_i \tag{15}$$

ideal. We prove that given ideal ranks, inequality 15 holds.

Assume $\forall i \;\; \hat{R}_i \leq R_i$, then:

$$\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} \leq \sum_{i=1}^{N} R_i \times I_i \qquad \forall i \;\; L_i \geq 1 \tag{16}$$

Consider α as:

$$\begin{aligned} \alpha &= \sum_{i=1}^{N} R_i \times I_i - \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} \\ &= \sum_{i=1}^{N} (R_i - \frac{\hat{R}_i}{L_i}) I_i \end{aligned} \tag{17}$$

Then as long as the first part of equation 13 is less than α, the inequality 15 holds.

$$\sum_{i=0}^{N} r_i \times L_i \leq \alpha \tag{18}$$

Given the assumption and the inequality 18:

$$\begin{aligned}\sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} &\leq \alpha + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} \\ &\leq \sum_{i=1}^{N} R_i \times I_i - \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} \\ &\leq \sum_{i=1}^{N} R_i \times I_i \end{aligned} \tag{19}$$

Therefore, with ideal ranks, $\forall i \;\; \hat{R}_i \leq R_i$ and $\sum_{i=0}^{N} r_i \times L_i \leq \alpha$, inequality 15 holds. □

Given no ideal, or some ideal ranks, there is no guarantee that inequality 15 holds without further constraints. We will briefly discuss some ideal settings for semi ideal ranks.

semi ideal. We prove that given semi ideal ranks, inequality 15 holds.

Assume $\forall i \;\; \hat{R}_i > R_i$, then:

$$k_i = \hat{R}_i - R_i \qquad \forall i \tag{20}$$

We can consider:

$$\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} = \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} + \sum_{i=1}^{N} k_i \times \frac{I_i}{L_i} \tag{21}$$

Consider β as:

$$\begin{aligned}\beta &= \sum_{i=1}^{N} R_i \times I_i - \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} \\ &= \sum_{i=1}^{N} R_i \times I_i \times (1 - \frac{1}{L_i}) \end{aligned} \tag{22}$$

Then as long as the second part of equations 21 is less than β, the inequality 16 holds.

$$\sum_{i=1}^{N} k_i \times \frac{I_i}{L_i} \leq \beta \tag{23}$$

Given the assumption and the inequality 23:

$$\begin{aligned}\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} &= \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} + \sum_{i=1}^{N} k_i \times \frac{I_i}{L_i} \\ &\leq \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} + \beta \\ &\leq \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} + \sum_{i=1}^{N} R_i \times I_i - \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} \\ &\leq \sum_{i=1}^{N} R_i \times I_i \end{aligned} \tag{24}$$

Therefore, with ideal splits and $\sum_{i=1}^{N} k_i \times \frac{I_i}{L_i} \leq \beta$, inequality 16 holds. The rest of the proof is similar to ***ideal*** case. □

We will now prove that as long as the only existed semi-ideal ranks, follow the rule above, inequality 15 can hold.

semi ideal. We prove that given just some ideal ranks, inequality 15 holds.

Assume $\exists j \;\; \hat{R}_j > R_j$, then:

$$k_j = \hat{R}_j - R_j \qquad \forall j \;\; s.t \;\; \hat{R}_j > R_j \tag{25}$$

We can consider $N_1 = \left\{\forall i \mid \hat{R}_i \leq i\right\}$ and $N_2 = \left\{\forall j \mid \hat{R}_j > j\right\}$:

$$\begin{aligned}
\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} &= \sum_{i \in N_1} \hat{R}_i \times \frac{I_i}{L_i} + \sum_{j \in N_2} \hat{R}_j \times \frac{I_j}{L_j} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} \hat{R}_j \times \frac{I_j}{L_j} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \sum_{j \in N_2} k_j \times \frac{I_j}{L_j}
\end{aligned} \tag{26}$$

Consider β as:

$$\begin{aligned}
\beta &= \sum_{i=1}^{N} R_i \times I_i - \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} \\
&= \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times I_j - \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_i}{L_i} \\
&= \sum_{j \in N_2} R_j \times I_j - \sum_{j \in N_2} R_j \times \frac{I_i}{L_i} \\
&= \sum_{j \in N_2} R_j \times I_j \times (1 - \frac{1}{L_j})
\end{aligned} \tag{27}$$

Then as long as the last part of inequality 26 is less than β, the inequality 16 holds.

$$\sum_{j \in N_2} k_j \times \frac{I_j}{L_j} \leq \beta \tag{28}$$

Given the assumption and the inequality 28:

$$\begin{aligned}
\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} &\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \sum_{j \in N_2} k_j \times \frac{I_j}{L_j} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \beta \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \sum_{j \in N_2} R_j \times I_j - \sum_{j \in N_2} R_j \times \frac{I_i}{L_i} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times I_j \\
&\leq \sum_{i=1}^{N} R_i \times I_i
\end{aligned} \tag{29}$$

Therefore, with ideal splits as long as the existed semi ideal ranks follow inequality 28, inequality 16 holds. The rest of the proof is similar to ***ideal*** case. □

B Method analysis on TRL

As stated in subsection 4.3, equation 14, the number of required parameters for a TRL on our framework is ideally less than equations 10. Given $\mathcal{T} \in \mathbb{R}^{S_0, I_1 \times \cdots \times I_N}$

and $\hat{\mathcal{T}} \in \mathbb{R}^{S_0, L_1 \times \cdots \times L_N \times \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}}$, and output $\mathcal{Y} \in \mathbb{R}^{S_0 \times O}$, ideally a rank $(r_1, \ldots, r_N, \hat{R}_1, \ldots, \hat{R}_N, \hat{R}_{N+1})$ TRL on $\hat{\mathcal{T}}$ has less parameters than a rank $(R_1, \ldots, R_N, R_{N+1})$ TRL on $\mathcal{T}$. Since TRL ranks are arbitrary values, one can choose these ranks in a way that inequality 30 does not hold.

$$\prod_{i=1}^{N} r_i \prod_{i=1}^{N+1} \hat{R}_i + \sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \leq \prod_{i=1}^{N+1} R_i + \sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O \tag{30}$$

ideal. We prove that given ideal ranks, inequality 30 holds.

Assume $\forall i \;\; \hat{R}_i \leq R_i$, then:

$$\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \leq \sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O \qquad \forall i \;\; L_i \geq 1 \tag{31}$$

Consider α_1 as:

$$\begin{aligned} \alpha_1 &= \sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O - \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\ &= \sum_{i=1}^{N} (R_i - \frac{\hat{R}_i}{L_i}) \times I_i + (R_{N+1} - \hat{R}_{N+1}) \times O \end{aligned} \tag{32}$$

Then as long as the second part of equation 14 is less than α_1,

$$\sum_{i=1}^{N} L_i \times r_i \leq \alpha_1 \tag{33}$$

It proves that :

$$\begin{aligned} &\sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\ &\leq \alpha_1 + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\ &\leq \sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O - \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\ &\leq \sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O \end{aligned} \tag{34}$$

As long as $\prod_{i=1}^{N} r_i \prod_{i=1} \hat{R}_i \leq \prod_{i=1}^{N+1} R_i$, inequality 30 holds using 34.

$$\begin{aligned} &\prod_{i=1}^{N} r_i \prod_{i=1}^{N+1} \hat{R}_i + \sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\ &\leq \prod_{i=1}^{N+1} R_i + \sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\ &\leq \prod_{i=1}^{N+1} R_i + \sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O \end{aligned} \tag{35}$$

Assuming $\exists i \;\; \prod_{i=1}^{N} r_i \prod_{i=1} \hat{R}_i > \prod_{i=1}^{N+1} R_i$, then consider α_2 as:

$$\alpha_2 = \frac{\prod_{i=1}^{N+1} R_i}{\prod_{i=1}^{N+1} \hat{R}_i} \tag{36}$$

As long as $\prod_{i=1}^{N} r_i \leq \alpha_2$, inequality 30 holds using 34. However, if $\prod_{i=1}^{N} r_i > \alpha_2$, then consider k as:

$$k = \prod_{i=1}^{N} r_i \prod_{i=1} \hat{R}_i - \prod_{i=1}^{N+1} R_i \tag{37}$$

Given:

$$k \leq \left(\sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O \right) - \left(\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \right) \tag{38}$$

Therefore, with ideal ranks, $\forall i \;\; \hat{R}_i \leq R_i$, $\sum_{i=0}^{N} r_i L_i \leq \alpha_1$, and $\prod_{i=1}^{N} r_i \prod_{i=1} \hat{R}_i \leq \prod_{i=1}^{N+1} R_i$, or $\prod_{i=1}^{N} r_i \leq \alpha_2$, and inequality 38, inequality 30 holds. □

Given no ideal, or some ideal ranks, there is no guarantee that inequality 30 holds without further constraints. We will briefly discuss some ideal settings for semi ideal ranks and splits.

semi ideal. We prove that given semi ideal ranks, inequality 30 holds.

Assume $\forall i \;\; \hat{R}_i > R_i$, which is equivalent to solving for $\exists j \;\; \hat{R}_j > R_j$, then:

$$k_i = \hat{R}_i - R_i \tag{39}$$

We can consider:

$$\sum_{i=1}^{N} \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O = \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} + \sum_{i=1}^{N} k_i \times \frac{I_i}{L_i} + R_{N+1} \times O + k_{N+1} \times O \tag{40}$$

Similar to TCL, consider β as:

$$\begin{aligned} \beta &= \sum_{i=1}^{N} R_i \times I_i - \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} \\ &= \sum_{i=1}^{N} R_i \times I_i \times (1 - \frac{1}{L_i}) \end{aligned} \tag{41}$$

Then as long as the k_i parts of equation 40 is less than β, the inequality 31 holds.

$$\sum_{i=1}^{N} k_i \times \frac{I_i}{L_i} + k_{N+1} \times O \leq \beta \tag{42}$$

Therefore, given the assumption and the inequality 42, with ideal splits, 31 holds. The rest of the proof is similar to ***ideal*** case. □

C Subspace of MLP : TCL

We will discuss the subspace attribute of our method on TCL, classic TCL, and MLP. Given input tensor $\mathcal{X} \in \mathbb{R}^{S_0 \times I_1 \times \cdots \times I_N}$, $\mathcal{X}_{[0]}$ is the vectorization of all samples within the batch S_0.

Consider:

$$\bar{\mathcal{X}} = \mathcal{X}_{[0]} \in \mathbb{R}^{S_0 \times I} \qquad I = I_1 I_2 \dots I_N \tag{43}$$

A MLP can be represented as:

$$\mathcal{Y} = \bar{\mathcal{X}} V^T \qquad V \in \mathbb{R}^{R \times I} \;\; , \;\; R = R_1 R_2 \dots R_N \tag{44}$$

Consider the solution space Φ such that:

$$\Phi = \left\{ \bar{\mathcal{X}} V^T | V \in \mathbb{R}^{R \times I} \right\} \tag{45}$$

Assuming $V^{(i)} \in \mathbb{R}^{R_i \times I_i}$, then a TCL with factors $V^{(i)}$ can be defined as 7:

$$\begin{aligned}\hat{\mathcal{X}}_{[0]} &= \mathcal{X}_{[0]}(V^{(1)} \otimes \cdots \otimes V^{(N)})^T \\ &= \bar{\mathcal{X}}\hat{V}^T\end{aligned} \tag{46}$$

where $\hat{V} = \left(V^{(1)} \otimes V^{(2)} \otimes \cdots \otimes V^{(N)}\right) \in \mathbb{R}^{R \times I}$.

Consider the solution space Ψ such that:

$$\Psi = \left\{\bar{\mathcal{X}}\hat{V}^T | \hat{V} = \left(V^{(1)} \otimes \cdots \otimes V^{(N)}\right) \in \mathbb{R}^{R \times I} \wedge \forall i \;\; V^{(i)} \in \mathbb{R}^{R_i \times I_i}\right\} \tag{47}$$

It is clear that $\Psi \subseteq \Phi$. So the solution of a TCL is a subset of MLP but with less parameters.

Let Γ be a reshape function introduced in section 4:

$$\tilde{\mathcal{X}} = \Gamma(\mathcal{X}) \in \mathbb{R}^{L_1 \times \cdots \times L_N, \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}} \tag{48}$$

Consider $U^{(i)} \in \mathbb{R}^{R_i \times \frac{I_i}{L_i}}$ and $W^{(i)} \in \mathbb{R}^{r_i \times L_i}$, then a TCL with factors $U^{(i)}$ and $W^{(i)}$ can be defined as :

$$\begin{aligned}\hat{\mathcal{X}} &= \left(W^{(1)}, \ldots, W^{(N)}, U^{(1)}, \ldots, U^{(N)}\right)_{1:2N} \tilde{\mathcal{X}} \\ &= \left(W^{(1)} \otimes U^{(1)}, \ldots, W^{(N)} \otimes U^{(N)}\right)_{1:N} \Gamma^{-1}(\tilde{\mathcal{X}}) \\ &= \left(V^{(1)}, \ldots, V^{(N)}\right)_{1:N} \mathcal{X} \\ &= \bar{\mathcal{X}}\tilde{V}^T\end{aligned} \tag{49}$$

Therefore, similar to equation 7 :

$$\hat{\mathcal{X}}_{[0]} = \bar{\mathcal{X}} \times (\tilde{V})^T \tag{50}$$

where Γ^{-1} is the inverse of the reshape operation; $\tilde{V} = \left(V^{(1)} \otimes V^{(2)} \otimes \cdots \otimes V^{(N)}\right) \in \mathbb{R}^{R' \times I}$, $V^{(i)} = W^{(i)} \otimes U^{(i)}$, and $R' = \prod_{i=1}^{N} R_i r_i$

Consider the solution space Ω such that:

$$\begin{aligned}\Omega = \Big\{\bar{\mathcal{X}}\tilde{V}^T | \tilde{V} = \left(V^{(1)} \otimes V^{(2)} \otimes \cdots \otimes V^{(N)}\right) \in \mathbb{R}^{R' \times I}\Big\} \\ \textbf{s.t.} \\ \forall i \;\; V^{(i)} = W^{(i)} \otimes U^{(i)} \in \mathbb{R}^{R_i r_i \times I_i} \\ \forall i \;\; U^{(i)} \in \mathbb{R}^{R_i \times \frac{I_i}{L_i}}, W^{(i)} \in \mathbb{R}^{r_i \times L_i}\end{aligned} \tag{51}$$

It is clear that $\Omega \subseteq \Psi \subseteq \Phi$. So the solution of a TCL on our framework is a subset of standard TCL and MLP but with less parameters.

D Subspace of MLP : TRL

We will discuss the subspace attribute of our methods on TRL, classic TRL, and MLP. Given input tensor $\mathcal{X} \in \mathbb{R}^{S_0 \times I_1 \times \cdots \times I_N}$, similar to TCL C :

$$\bar{\mathcal{X}} = \mathcal{X}_{[0]} \in \mathbb{R}^{S_0 \times I} \qquad I = I_1 I_2 \ldots I_N \tag{52}$$

A MLP can be represented as:

$$\mathcal{Y} = \langle \mathcal{X}, \mathcal{W} \rangle_N \tag{53}$$

where $\mathcal{Y} \in \mathbb{R}^{S_0 \times O}$ and $\mathcal{W} \in \mathbb{R}^{I_1 \times \cdots \times I_N \times O}$.

We can write equations 53 as :

$$\mathcal{Y} = \bar{\mathcal{X}}\bar{\mathcal{W}}^T \tag{54}$$

where $\bar{\mathcal{W}} = \mathcal{W}_{[N+1]} \in \mathbb{R}^{O \times I}$.

Consider solution space Φ' such that:

$$\Phi' = \left\{ \bar{\mathcal{X}} \bar{\mathcal{W}}^T | \bar{\mathcal{W}} \in \mathbb{R}^{O \times I} \right\} \tag{55}$$

Assume $V^{(i)} \in \mathbb{R}^{R_i \times I_i}$, $V^{(N+1)} \in \mathbb{R}^{R_{N+1} \times O}$, and core tensor $\mathcal{G} \in \mathbb{R}^{R_1 \times R_{N+1}}$, then a TRL with factors $V^{(i)}$ and core $\mathcal{G}$ can be defined as:

$$\begin{aligned}
\mathcal{Y} &= \langle \mathcal{X}, \mathcal{W} \rangle_N \\
&= \left\langle \mathcal{X}, \left(V^{(1)}, \ldots, V^{(N+1)} \right)_{0:N+1} \mathcal{G} \right\rangle_N \\
&= \left\langle \left(V^{(1)^T}, \ldots, V^{(N)^T} \right)_{1:N} . \mathcal{X}, \left(V^{(N+1)} \right)_{N+1} \mathcal{G} \right\rangle_N \\
&= \left\langle \hat{\mathcal{X}}, \hat{\mathcal{G}} \right\rangle_N
\end{aligned} \tag{56}$$

Similar to TCL 45, consider :

$$\bar{\hat{\mathcal{X}}} = \hat{\mathcal{X}}_{[0]} \in \mathbb{R}^{S_0 \times R} \qquad \textbf{and} \qquad \bar{\hat{\mathcal{G}}} = \hat{\mathcal{G}}_{[N+1]} \in \mathbb{R}^{O \times R} \tag{57}$$

Where $R = R_1 R_2 \ldots R_N$, we can consider the TRL as :

$$\mathcal{Y} = \bar{\hat{\mathcal{X}}} \bar{\hat{\mathcal{G}}}^T \in \mathbb{R}^{S_0 \times O} \tag{58}$$

Consider solution space Ψ' such that:

$$\Psi' = \left\{ \bar{\hat{\mathcal{X}}} \bar{\hat{\mathcal{G}}}^T | \text{ conditions 57, 56 holds } \right\} \tag{59}$$

It is clear that $\Psi' \subseteq \Phi'$. So the solution of a TRL is a subset of MLP but with less parameters.

Let Γ be a reshape function introduced in section 4:

$$\tilde{\mathcal{X}} = \Gamma(\mathcal{X}) \in \mathbb{R}^{L_1 \times \cdots \times L_N, \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}} \tag{60}$$

Consider $U^{(i)} \in \mathbb{R}^{R_i \times \frac{I_i}{L_i}}$, $U^{(N+1)} \in \mathbb{R}^{R_{N+1} \times O}$, $W^{(i)} \in \mathbb{R}^{r_i \times L_i}$, and core tensor $\mathcal{G} \in \mathbb{R}^{r_1 \times \cdots \times r_N \times R_1 \times \cdots \times R_N \times R_{N+1}}$, then a TRL with factors $U^{(i)}$, $W^{(i)}$, and core $\mathcal{G}$ can be defined as :

$$\begin{aligned}
\mathcal{Y} &= \left\langle \tilde{\mathcal{X}}, \mathcal{W} \right\rangle_N \\
&= \left\langle \tilde{\mathcal{X}}, \left(W^{(1)}, \ldots, W^{(N)}, U^{(1)}, \ldots, U^{(N+1)} \right)_{0:2N+1} . \mathcal{G} \right\rangle_N \\
&= \left\langle \left(W^{(1)^T}, \ldots, W^{(N)^T}, U^{(1)^T}, \ldots, U^{(N)^T} \right)_{1:2N} . \tilde{\mathcal{X}}, \left(U^{(N+1)} \right)_{N+1} . \mathcal{G} \right\rangle_N \\
&= \left\langle \left(W^{(1)^T} \otimes U^{(1)^T}, \ldots, W^{(N)^T} \otimes U^{(N)^T} \right)_{1:N} . \Gamma^{-1}(\tilde{\mathcal{X}}), \left(U^{(N+1)} \right)_{N+1} . \mathcal{G} \right\rangle_N \\
&= \left\langle \left(\tilde{V}^{(1)}, \ldots, \tilde{V}^{(N)} \right)_{1:N} . \Gamma^{-1}(\tilde{\mathcal{X}}), \left(\tilde{V}^{(N+1)} \right)_{N+1} . \mathcal{G} \right\rangle_N \\
&= \left\langle \hat{\tilde{\mathcal{X}}}, \hat{\tilde{\mathcal{G}}} \right\rangle_N
\end{aligned} \tag{61}$$

which is deduced from equation 56. Similar to previous section C, consider:

$$\bar{\hat{\tilde{\mathcal{X}}}} = \hat{\tilde{\mathcal{X}}}_{[0]} \in \mathbb{R}^{S_0 \times R'} \qquad \textbf{and} \qquad \bar{\hat{\tilde{\mathcal{G}}}} = \hat{\tilde{\mathcal{G}}}_{[N+1]} \in \mathbb{R}^{O \times R'} \tag{62}$$

where $R' = \prod_{i=1} R_i r_i$, we can consider the TRL as:

$$\mathcal{Y} = \bar{\hat{\tilde{\mathcal{X}}}} \bar{\hat{\tilde{\mathcal{G}}}}^T \in \mathbb{R}^{S_0 \times O} \tag{63}$$

Consider solution space Ω' such that:

$$\Omega' = \left\{ \bar{\hat{\tilde{\mathcal{X}}}} \bar{\hat{\tilde{\mathcal{G}}}}^T | \textbf{conditions 62, 61} \right\} \tag{64}$$

It is clear that $\Omega' \subseteq \Psi' \subseteq \Phi'$. So the solution of a TRL on our framework is a subset of standard TRL and MLP but with less parameters.

Table 4: MNIST applied to resnet50

TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline		737,290	97.25	99.93
$(200, 3, 3, 8)$		424,116	98.75	99.98
$(100, 3, 3, 10)$		213,936	97.99	99.96
$(50, 3, 3, 3)$		103,816	96.49	99.78
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^A$	103,512	97.81	99.95
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^C$	103,512	98.18	99.98
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^A$	20,612	98.56	99.94
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^C$	20,612	97.79	99.95
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^A$	3,534	97.85	99.96
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^C$	3,534	98.12	100.00
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^A$	1,386	97.74	99.93
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^C$	1,386	98.43	99.89

Table 5: CIFAR10 applied to ResNet50

TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline		737,290	90.71	99.71
$(200, 3, 3, 8)$		424,116	91.02	99.77
$(100, 3, 3, 10)$		213,936	92.31	99.70
$(50, 3, 3, 3)$		103,816	90.41	99.28
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^A$	103,512	91.78	99.79
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^C$	103,512	92.58	99.83
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^A$	20,612	90.99	99.46
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^C$	20,612	91.52	99.50
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^A$	3,534	92.26	99.82
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^C$	3,534	91.94	99.83
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^A$	1,386	90.44	99.32
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^C$	1,386	89.03	99.43

E Additional tables

Additional experiments evaluated on MNIST, CIFAR10, and Tiny ImageNet-200 applied to ResNet50, ResNet101, VGG-19 , and classic CNN. In our experiments, to better represent the effect of our methods, we ignored adaptive average pooling layer of all models.

Table 6: Tiny ImageNet-200 applied to ResNet50

TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline		14,745,800	75.77	90.82
$(1024, 3, 3, 200)$		3,980,388	80.68	93.72
$(600, 3, 3, 200)$		2,348,836	81.08	93.80
$(500, 3, 3, 150)$		1,729,036	81.33	94.00
$(1, 1, 1, 500, 1, 1, 200)$	$(2, 2, 2)^A$	652,012	81.86	94.57
$(1, 1, 1, 500, 1, 1, 200)$	$(2, 2, 2)^C$	652,012	81.94	94.32
$(1, 1, 1, 100, 1, 1, 100)$	$(2, 2, 2)^A$	132,412	82.02	94.54
$(1, 1, 1, 100, 1, 1, 100)$	$(2, 2, 2)^C$	132,412	82.73	94.49
$(4, 1, 1, 100, 1, 1, 200)$	$(16, 2, 2)^A$	222,474	82.64	94.87
$(4, 1, 1, 100, 1, 1, 200)$	$(16, 2, 2)^C$	222,474	81.92	94.62
$(4, 1, 1, 50, 1, 1, 100)$	$(16, 2, 2)^A$	91,274	82.38	94.35
$(4, 1, 1, 50, 1, 1, 100)$	$(16, 2, 2)^C$	91,274	82.30	94.47

Table 7: MNIST applied to ResNet101

TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline		737,290	98.58	99.97
$(200, 3, 3, 8)$		424,116	98.35	99.98
$(100, 3, 3, 10)$		213,936	97.99	99.98
$(50, 3, 3, 3)$		103,816	97.42	99.86
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^A$	103,512	98.03	99.98
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^C$	103,512	97.77	99.96
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^A$	20,612	98.24	99.84
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^C$	20,612	98.24	99.90
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^A$	3,534	98.40	100
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^C$	3,534	98.25	99.95
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^A$	1,386	98.03	99.96
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^C$	1,386	98.05	99.89

Table 8: CIFAR10 applied to ResNet101

TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline		737,290	92.67	99.71
$(200, 3, 3, 8)$		424,116	91.96	99.78
$(100, 3, 3, 10)$		213,936	90.66	99.83
$(50, 3, 3, 3)$		103,816	90.30	99.19
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^A$	103,512	92.18	99.79
$(1, 1, 1, 100, 1, 1, 10)$	$(2, 2, 2)^C$	103,512	91.85	99.88
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^A$	20,612	92.07	99.61
$(1, 1, 1, 20, 1, 1, 4)$	$(2, 2, 2)^C$	20,612	90.28	99.29
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^A$	3,534	90.34	99.78
$(4, 1, 1, 20, 1, 1, 10)$	$(16, 2, 2)^C$	3,534	92.14	99.87
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^A$	1,386	92.43	99.64
$(4, 1, 1, 10, 1, 1, 4)$	$(16, 2, 2)^C$	1,386	89.96	99.26

Table 9: Tiny ImageNet-200 applied to ResNet101

TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline		14,745,800	79.81	92.28
$(1024, 3, 3, 200)$		3,980,388	84.10	94.55
$(600, 3, 3, 200)$		2,348,836	84.20	94.98
$(500, 3, 3, 150)$		1,729,036	84.55	94.99
$(1, 1, 1, 500, 1, 1, 200)$	$(2, 2, 2)^A$	652,012	84.99	95.49
$(1, 1, 1, 500, 1, 1, 200)$	$(2, 2, 2)^C$	652,012	85.52	95.82
$(1, 1, 1, 100, 1, 1, 100)$	$(2, 2, 2)^A$	132,412	84.86	95.29
$(1, 1, 1, 100, 1, 1, 100)$	$(2, 2, 2)^C$	132,412	85.13	95.26
$(4, 1, 1, 100, 1, 1, 200)$	$(16, 2, 2)^A$	222,474	85.22	95.52
$(4, 1, 1, 100, 1, 1, 200)$	$(16, 2, 2)^C$	222,474	85.35	95.45
$(4, 1, 1, 50, 1, 1, 100)$	$(16, 2, 2)^A$	91,274	85.20	95.21
$(4, 1, 1, 50, 1, 1, 100)$	$(16, 2, 2)^C$	91,274	85.24	95.36

Table 10: CIFAR10 applied to classic CNN

TCL	TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline			4,729,866	76.09	98.04
$(128, 6, 6)$	$(128, 6, 6, 10)$		79,092	75.06	98.32
$(2, 1, 1, 64, 6, 6)$	$(2, 1, 1, 64, 6, 6, 10)$	$(2, 1, 1)^A$	54,528	73.57	98.09
$(2, 1, 1, 64, 6, 6)$	$(2, 1, 1, 64, 6, 6, 10)$	$(2, 1, 1)^C$	54,528	73.03	98.06

Table 11: Tiny ImageNet-200 applied to VGG-19

TCL	TRL	Split	Parameters	Top-1(%)	Top-5(%)
Baseline			93,102,280	63.66	84.16
$(512, 6, 6)$	$(512, 6, 6, 200)$		4,250,832	67.32	87.87
$(2, 2, 2, 256, 1, 1)$	$(1, 1, 1, 256, 1, 1, 200)$	$(2, 2, 2)^A$	666,714	68.91	88.16
$(2, 2, 2, 256, 1, 1)$	$(1, 1, 1, 256, 1, 1, 200)$	$(2, 2, 2)^C$	666,714	67.68	88.15
$(6, 2, 2, 64, 3, 3)$	$(4, 2, 2, 64, 1, 1, 200)$	$(8, 2, 2)^A$	253,104	68.66	88.33
$(6, 2, 2, 64, 3, 3)$	$(4, 2, 2, 64, 1, 1, 200)$	$(8, 2, 2)^C$	253,104	68.29	88.34

F Ablation studies

We generated synthetic samples from a Gaussian distribution $\mathcal{N}(0, 3)$ such that $\mathcal{X} \in \mathbb{R}^{S \times 32}$, where S is the number of samples. Given a fixed $\mathcal{W} \in \mathbb{R}^{32 \times 32}$, we consider $\mathcal{Y} = \mathcal{X} \times \mathcal{W}^T$. Note that $\mathcal{X}$ is the vectorization of each samples if they were higher order tensors and $\mathcal{Y} \in \mathbb{R}^{S \times 32}$. Our goal is to estimate $\mathcal{W}$ using our modules trained on noisy $\mathcal{X} + \epsilon$, where ϵ is from a Gaussian distribution $\mathcal{N}(0, 1)$. We evaluated our framework, standard TRL, and standard FC using RMSE; This study is carried out with a single layer model trained on noisy data using Adam optimizer. In figure 6, a comparison of different methods using same ranks is provided.

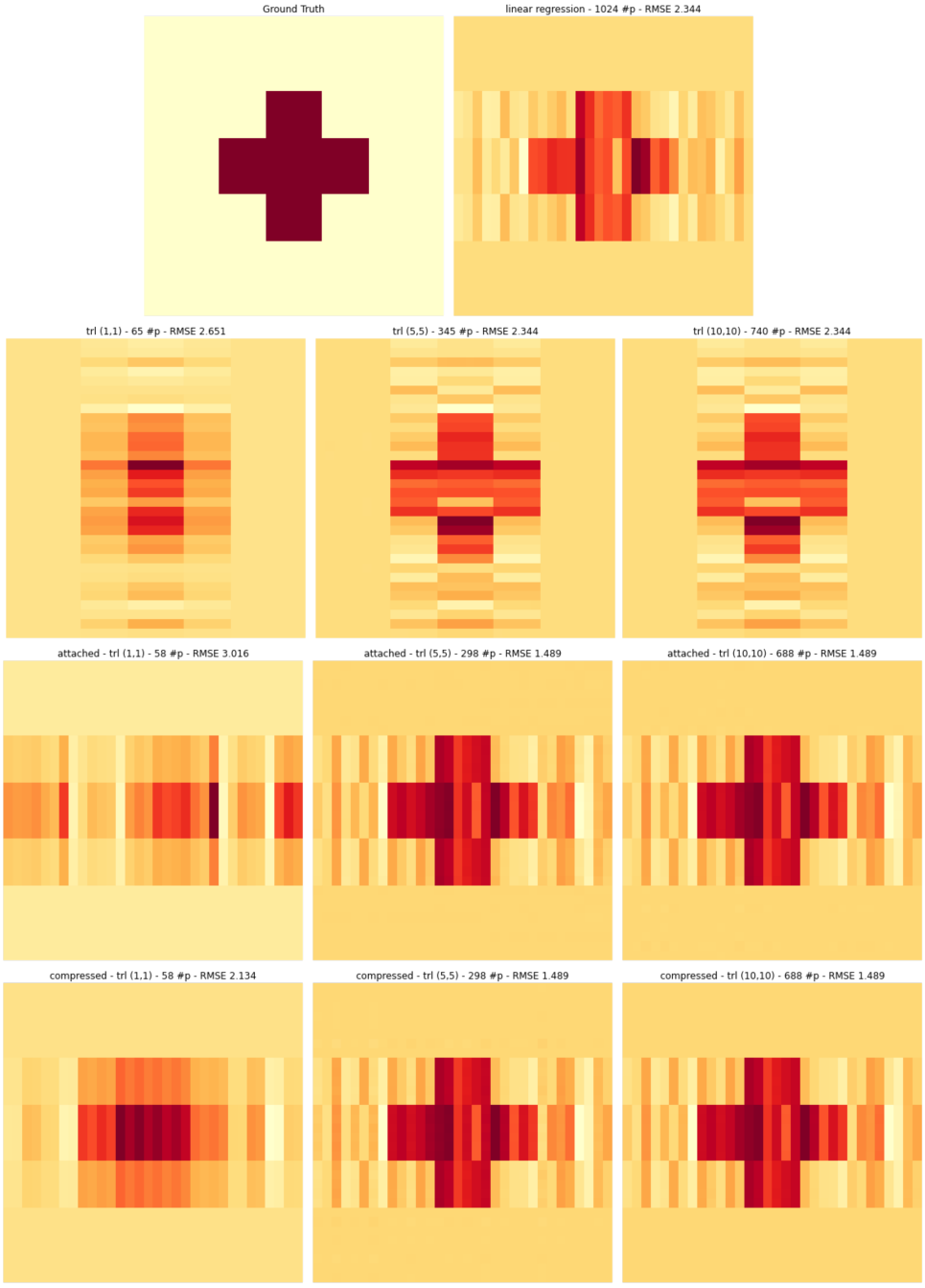

Figure 6: Our framework performs better with similar ranks compared to standard TRL or original FC. Ground Truth is the weight tensor (2-order tensor) displayed as an image, other modules tried to estimate $\mathcal{W}$ using noisy data.

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